



EE 551: Applied Photonics for Scientists and Engineers

Course Code	EE 551
Course Name	Applied Photonics for Scientists and Engineers
L-T-P-C	2-1-0-3
Prerequisites	IC110, IC111, IC221
Intended for	UG, PG, M. Tech, M. Sc. Physics, 1-PhD, PhD
Approval	recommended by the 38th BoA (Jan 2021)

Course Contents

- **Fundamentals of the Electromagnetic theory of light:** complex representation of electromagnetic field disturbances, Maxwell's equations, Fresnel equations, limitations of the electromagnetic description. [2 Lectures]
- **Optical systems:** Fermat's principle, basic optical elements, matrix methods for optics, thick lens and their systems, determination of cardinal points, basic optical systems. [3 Lectures]
- **Polarisation:** fundamentals, special devices - crystals, compensators, spatial light modulators, mathematical representations of polarised light. [3 Lectures]
- **Gaussian Beam Optics:** beam propagation equation, beam properties and their characterization, matrix approach for Gaussian beam optics. [2 Lectures]
- **Interferometry:** principles and applications - fundamental concepts - conditions for interference, coherence theory elements, Young's double slit experiment, multiple-beam interference. Systems - Michelson, Twyman Green, Fizeau and other select configurations. Selected applications- e.g. metrology, sensors. [4 Lectures]
- **Fourier Transforms in Optics:** Foundational concepts and theorems, Fourier methods in diffraction theory, Abbe-Porter's experiment, applications - e.g. optical waveshapers. [3 Lectures]
- **Fibre optic systems:** principles of guided wave propagation, basics of single mode and multimode, passive components, active components, fibre-optics based system design considerations, select applications - e.g. Dispersive Fourier Transformer, fibre optic sensors, imaging configurations. [4 Lectures]
- **Nonlinear optics:** Light-matter interaction and the nonlinear wave equation, second order nonlinearity- second harmonic generation, three-wave mixing, third order nonlinearity- third harmonic generation, four-wave mixing, Kerr nonlinearity and its applications. [4 Lectures]
- **Detection of Optical Radiation:** Time-domain methods: High speed detectors, Photomultipliers, Time of flight detectors, Cameras, characterization of ultrashort pulses. Spectral-domain methods -Essential components, resolving power of dispersive devices, the optical spectrum analyzer. Basics of quantum light -- single-photon generation and detection, applications - e.g. qubits. Full-field measurement techniques. [4 Lectures]

Tutorial sessions - 10.

Resource required for tutorial sessions - Access to computer labs for numerical/computational solving of problems.

Text Books

1. Hecht, E., **Optics**, 4th Edition, Pearson.
2. Ghatak, A. K., & Thyagarajan, K., **Optical Electronics**, Cambridge University Press, 2018.

References

1. Saleh, B. E. A., & Teich, M. C., **Fundamentals of Photonics**, 2nd Edition, Wiley Interscience 2007.